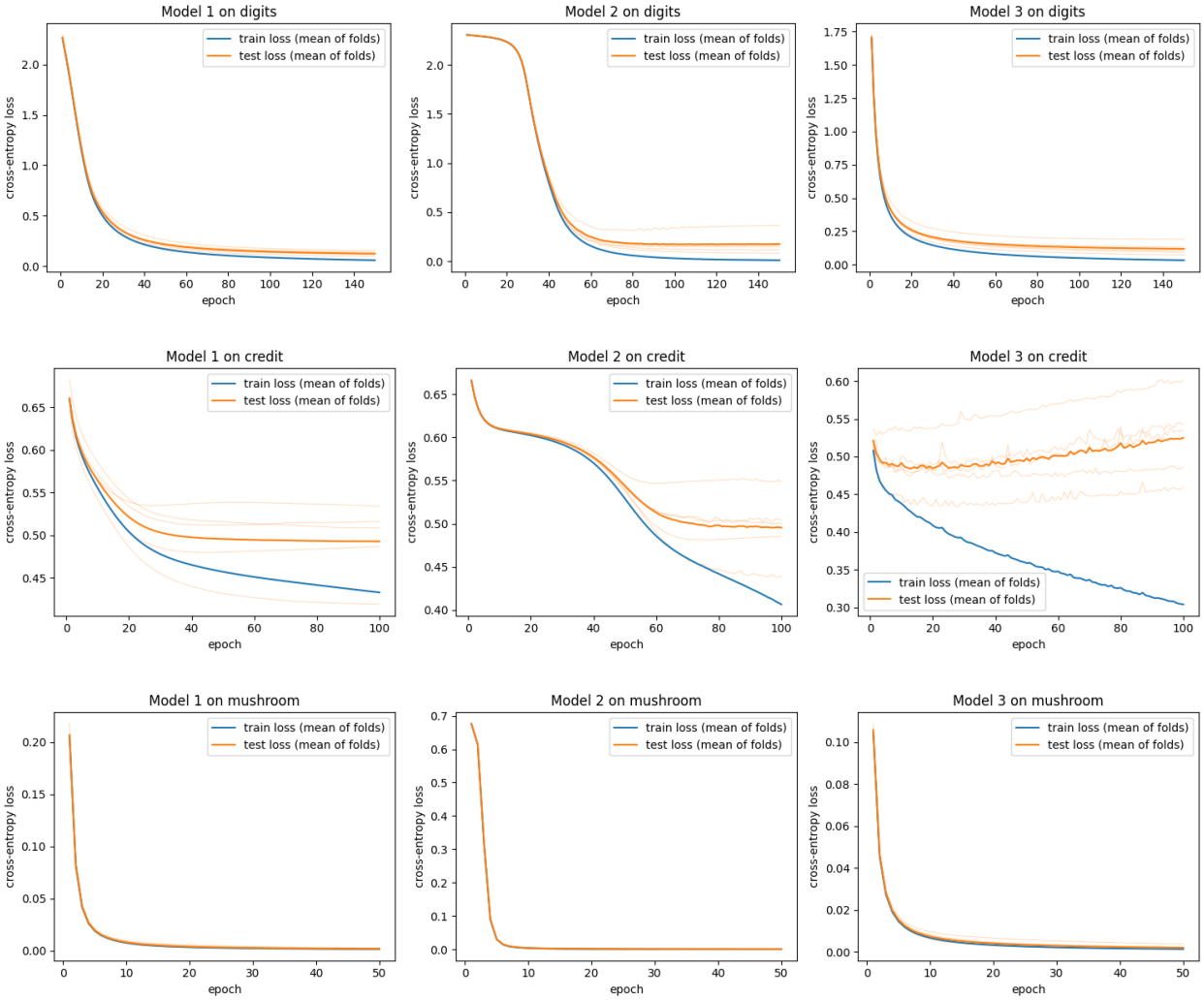
4.1



dataset	model	params	epochs	train loss	test loss	train acc	test acc	test std	seconds
digits	Model 1	1210	150	0.0584	0.1239	99.14%	96.72%	1.12%	18.5
digits	Model 2	37810	150	0.0103	0.1750	100.00%	96.44%	1.10%	38.7
digits	Model 3	37810	150	0.0329	0.1182	99.76%	97.39%	1.00%	27.9
credit	Model 1	370	100	0.4331	0.4920	79.38%	77.00%	2.26%	7.1
credit	Model 2	32602	100	0.4064	0.4954	81.85%	77.10%	1.56%	14.0
credit	Model 3	32593	100	0.3041	0.5240	87.55%	76.00%	3.46%	12.8
mushroom	Model 1	1906	50	0.0013	0.0016	100.00%	99.96%	0.05%	27.8
mushroom	Model 2	42202	50	0.0005	0.0007	100.00%	100.00%	0.00%	58.9
mushroom	Model 3	42247	50	0.0013	0.0020	100.00%	99.96%	0.05%	44.9

dataset	model	fold 1	fold 2	fold 3	fold 4	fold 5
digits	Model 1	96.94%	95.00%	96.10%	98.33%	97.21%
digits	Model 2	97.50%	94.72%	95.54%	97.21%	97.21%
digits	Model 3	97.78%	95.28%	97.77%	98.05%	98.05%
credit	Model 1	76.00%	76.00%	76.00%	81.50%	75.50%
credit	Model 2	77.50%	76.00%	76.00%	80.00%	76.00%
credit	Model 3	78.50%	74.50%	73.00%	81.50%	72.50%
mushroom	Model 1	100.00%	100.00%	100.00%	99.88%	99.94%
mushroom	Model 2	100.00%	100.00%	100.00%	100.00%	100.00%
mushroom	Model 3	100.00%	100.00%	100.00%	99.88%	99.94%

dataset	inputs	classes	L	model 1	model 2	model 3
digits	64	10	504	1210	37810	37810
credit	28	2	1417	370	32602	32593
mushroom	116	2	355	1906	42202	42247

The epochs that I chose were 150,100, and 50 for the respective digits, credit and mushroom datasets. While I feel like I could potentially improve the epoch counts a little more, for each of these it's an amount that for the most part gets the test/train values to converge without overfitting.

4.2

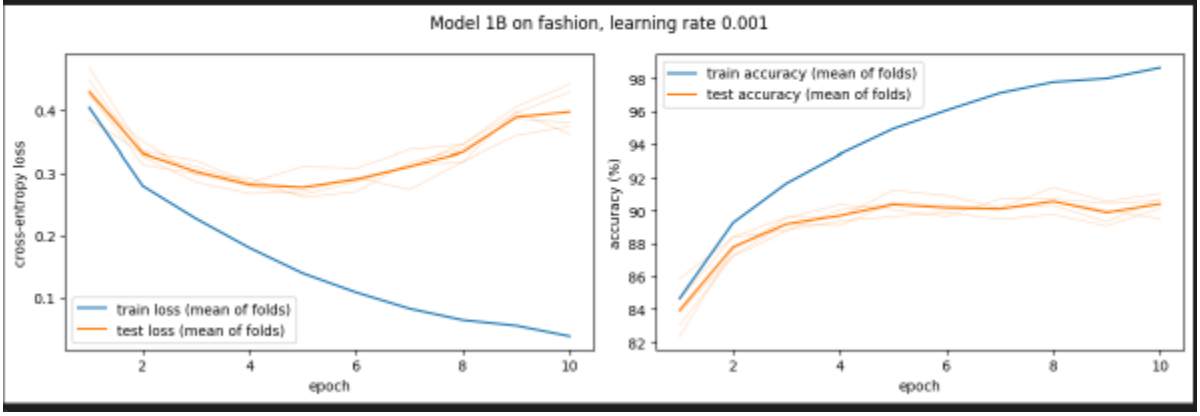
dataset	model	params	epochs	lr	s/fold	train acc	test loss	test acc	test std
fashion	1A	1633274	10	0.0010	7.1	96.66%	0.3193	89.81%	0.40%
fashion	1B	6692810	10	0.0010	25.7	98.66%	0.3974	90.41%	0.54%
fashion	2	6692810	15	0.0010	48.5	97.83%	0.2606	91.44%	0.55%
fashion	3	6693578	8	0.0010	24.7	97.55%	0.3013	90.47%	0.75%
fashion	4	4811786	15	0.0010	86.0	95.52%	0.2843	91.08%	0.60%
fashion	4	4811786	15	0.0100	85.0	87.45%	0.3886	86.37%	1.07%
digits	1A	1633274	10	0.0010	7.0	99.67%	0.0643	98.43%	0.23%
digits	1B	6692810	10	0.0010	26.9	99.72%	0.0660	98.51%	0.30%
digits	2	6692810	15	0.0010	40.4	99.78%	0.0443	98.83%	0.13%
digits	3	6693578	8	0.0010	24.6	99.41%	0.0757	98.48%	0.31%
digits	4	4811786	15	0.0010	85.3	99.38%	0.0571	98.67%	0.32%
digits	4	4811786	15	0.0100	84.9	97.87%	0.1070	97.30%	1.21%

Epochs: {"1A": 10, "1B": 10, "2": 15, "3": 8, "4": 15}

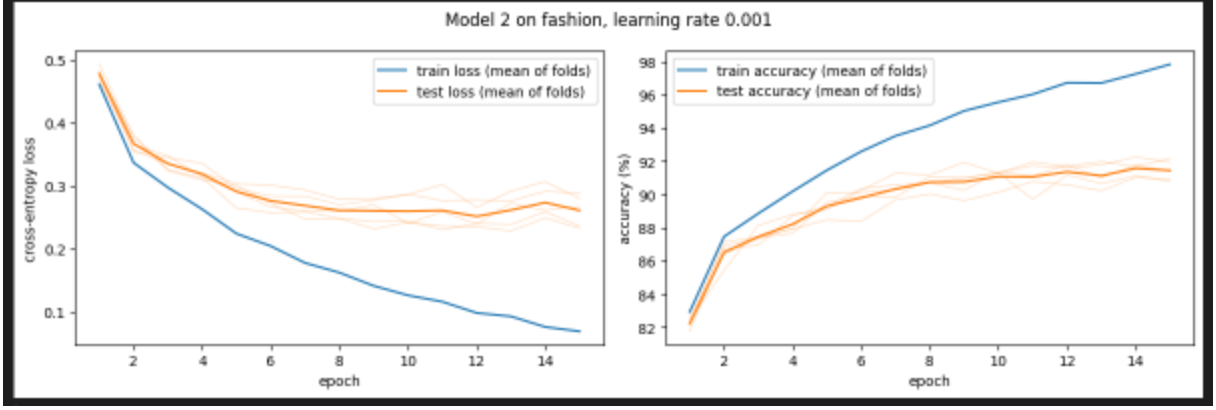
```
fold 1: train acc 96.36%  test acc 89.25%  9 s
fold 2: train acc 96.44%  test acc 89.72%  7 s
fold 3: train acc 96.27%  test acc 89.58%  7 s
fold 4: train acc 97.11%  test acc 90.10%  7 s
fold 5: train acc 97.11%  test acc 90.40%  7 s
```



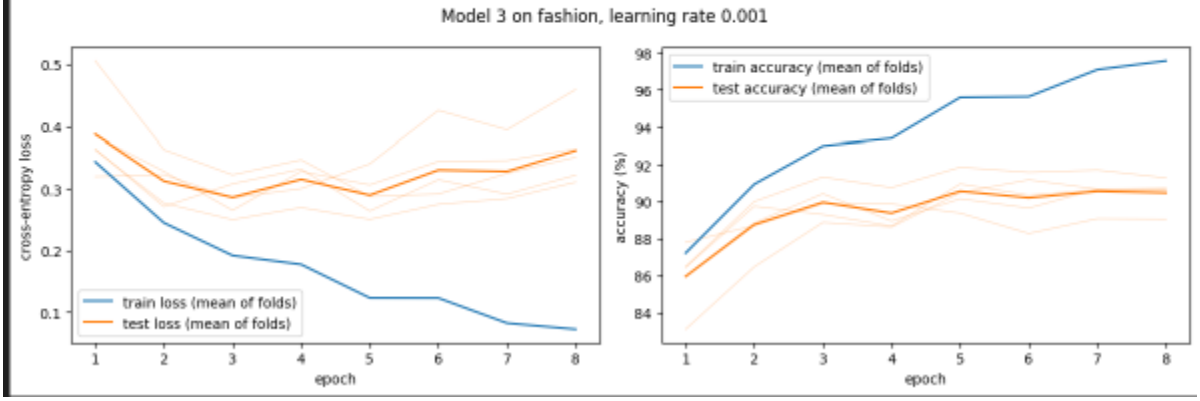

```
fold 1: train acc 98.81%  test acc 90.15%  25 s
fold 2: train acc 98.77%  test acc 91.05%  25 s
fold 3: train acc 97.92%  test acc 89.50%  26 s
fold 4: train acc 98.91%  test acc 90.75%  26 s
fold 5: train acc 98.87%  test acc 90.60%  26 s
```



```
fold 1: train acc 98.26%  test acc 90.92%  40 s
fold 2: train acc 97.42%  test acc 91.40%  40 s
fold 3: train acc 98.22%  test acc 90.77%  41 s
fold 4: train acc 98.26%  test acc 91.95%  41 s
fold 5: train acc 97.01%  test acc 92.17%  41 s
```

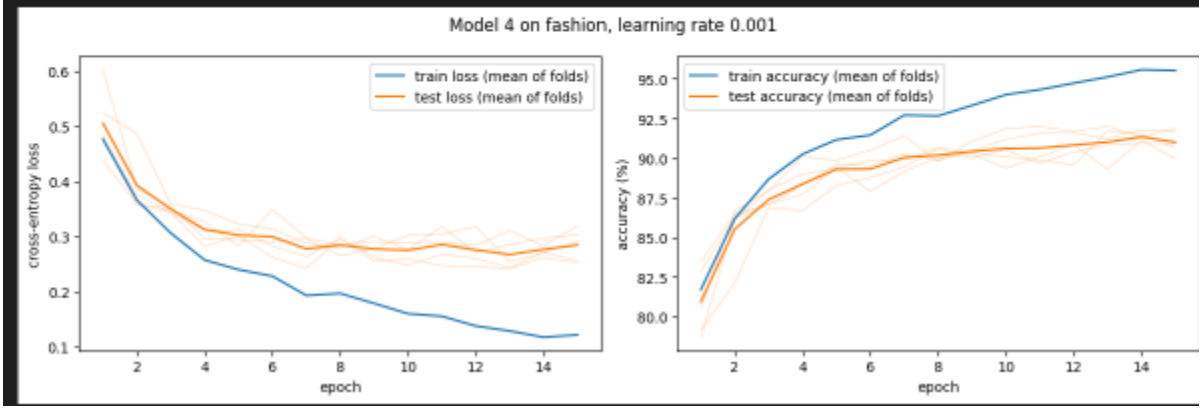



```
fold 1: train acc 97.32%  test acc 89.05%  25 s
fold 2: train acc 97.74%  test acc 90.70%  25 s
fold 3: train acc 97.57%  test acc 90.65%  25 s
fold 4: train acc 97.72%  test acc 90.70%  25 s
fold 5: train acc 97.42%  test acc 91.27%  25 s
```

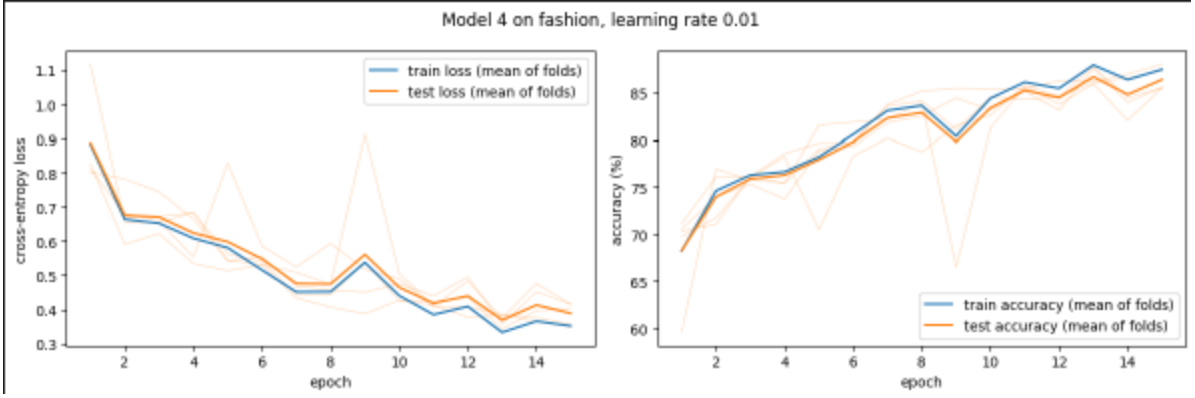



```
fold 2: train acc 95.69%  test acc 90.65%  86 s
fold 3: train acc 94.92%  test acc 90.88%  86 s
fold 4: train acc 96.00%  test acc 91.67%  86 s
fold 5: train acc 95.70%  test acc 91.85%  86 s
```

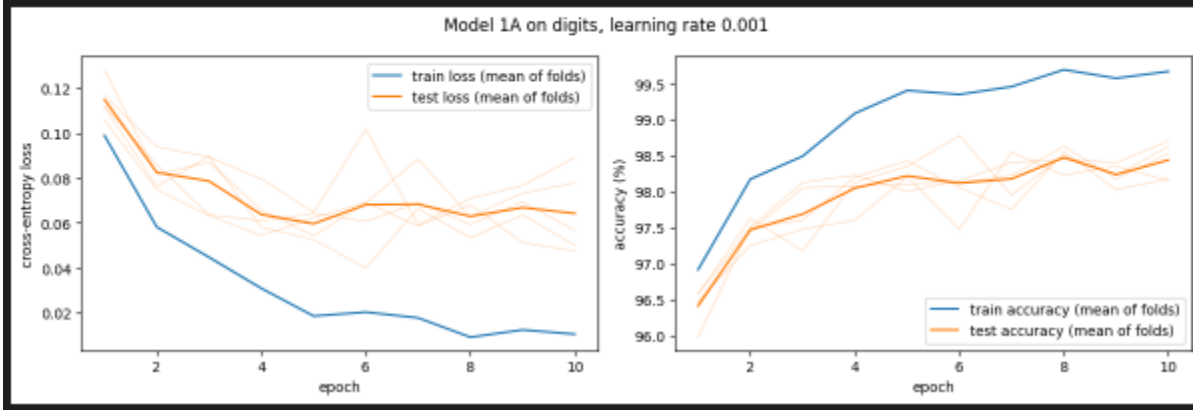
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...



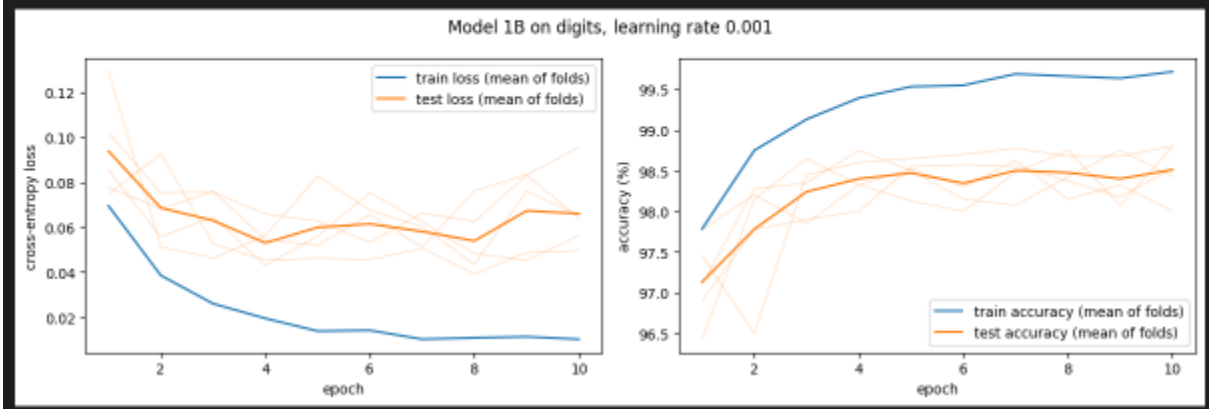
```
fold 1: train acc 90.19%  test acc 88.00%  85 s
fold 2: train acc 87.04%  test acc 85.52%  85 s
fold 3: train acc 86.09%  test acc 85.58%  85 s
fold 4: train acc 87.12%  test acc 85.45%  85 s
fold 5: train acc 86.81%  test acc 87.33%  85 s
```



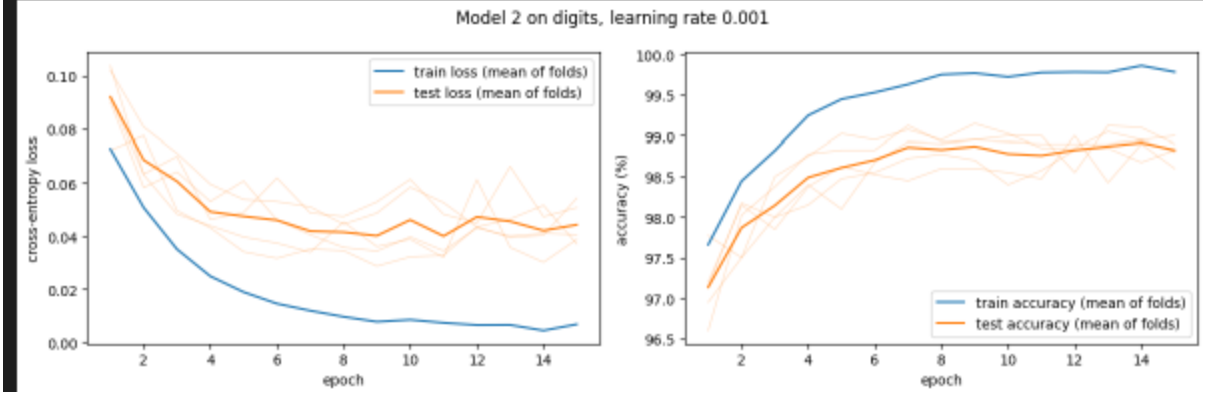

```
fold 1: train acc 99.54%  test acc 98.52%  7 s
fold 2: train acc 99.86%  test acc 98.70%  7 s
fold 3: train acc 99.52%  test acc 98.17%  7 s
fold 4: train acc 99.70%  test acc 98.62%  7 s
fold 5: train acc 99.61%  test acc 98.15%  7 s
```



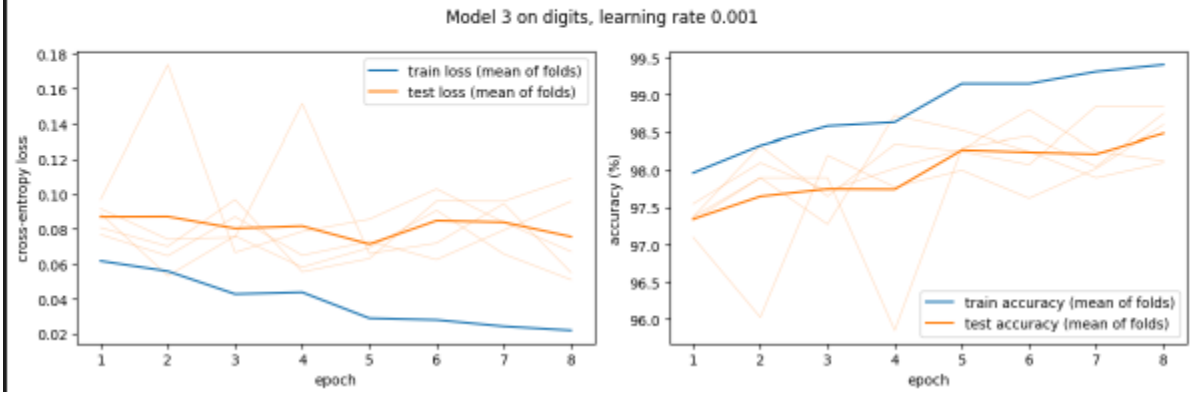

```
fold 1: train acc 99.76%  test acc 98.88%  27 s
fold 2: train acc 99.75%  test acc 98.42%  27 s
fold 3: train acc 99.52%  test acc 98.08%  27 s
fold 4: train acc 99.72%  test acc 98.88%  27 s
fold 5: train acc 99.84%  test acc 98.52%  27 s
```



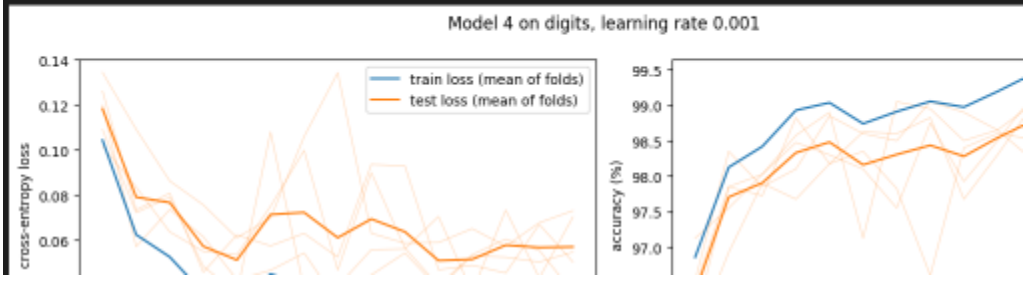
```
fold 1: train acc 99.81%  test acc 98.90%  41 s
fold 2: train acc 99.91%  test acc 99.00%  40 s
fold 3: train acc 99.61%  test acc 98.60%  40 s
fold 4: train acc 99.69%  test acc 98.80%  40 s
fold 5: train acc 99.89%  test acc 98.83%  40 s
```

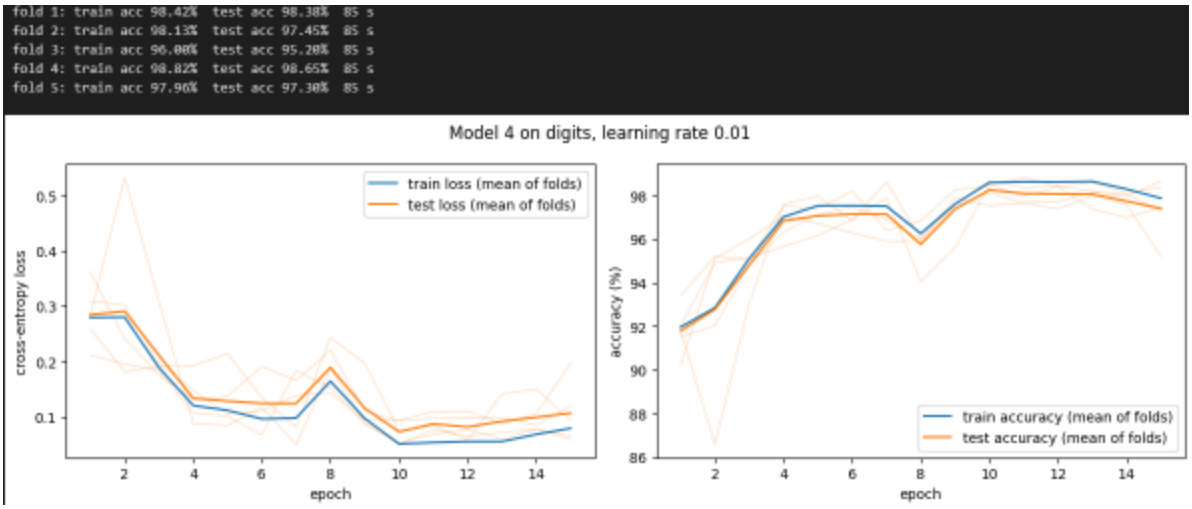



```
fold 1: train acc 99.66%  test acc 98.85%  25 s
fold 2: train acc 99.80%  test acc 98.10%  25 s
fold 3: train acc 99.46%  test acc 98.60%  25 s
fold 4: train acc 99.10%  test acc 98.12%  25 s
fold 5: train acc 99.74%  test acc 98.75%  25 s
```




```
fold 2: train acc 99.64%  test acc 99.02%  85 s
fold 3: train acc 99.48%  test acc 98.92%  85 s
fold 4: train acc 98.95%  test acc 98.45%  85 s
fold 5: train acc 99.41%  test acc 98.15%  85 s
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```





1A and 1B both diverge on the fashion sets, so they most likely need reduction in the amount of epochs. 1B boasts a marginally higher accuracy at 3-4x the training time, though that isn't a major issue with the current times. Model 3 has a very slight decrease in accuracy for similar training times while model 2 has a slight improvement in accuracy for double the training times of model 1B. Model 4 is double the training time of model 2 in both learning rates, with less training accuracy than any of the other models, though the decrease is still <10%. The higher learning rate version of model 4 has less accuracy than the low learning rate version, and the training/test losses and accuracy fit much better than all the other models. Because the model learned quicker, it's probably why the training data fit is so much better. It does diverge enough to not be overfitting.

In the digits dataset, 1B is marginally more accurate than 1A, by roughly .25%, with almost 4x the training time. Both could use a reduction in epoch. Model 2 and 3 and more accurate than 1B, with model 2 taking roughly 15 sec more and model 3 having comparable training times. Their models still have the inaccuracies that the previous models have in loss/accuracy comparisons. Once again, model 4 is more accurate than all the other models in test/train accuracy/loss. The low learning rate is similar to model 3 in accuracy; the higher rate lags a few percentage points behind. Both have roughly double the training time of model 2.

4.3

The trophy did not fit in the suitcase because it was too [MASK].

heavy	0.333
small	0.233
big	0.115

large 0.114

long 0.020

The [MASK] carefully cut the onion for dinner service tonight.

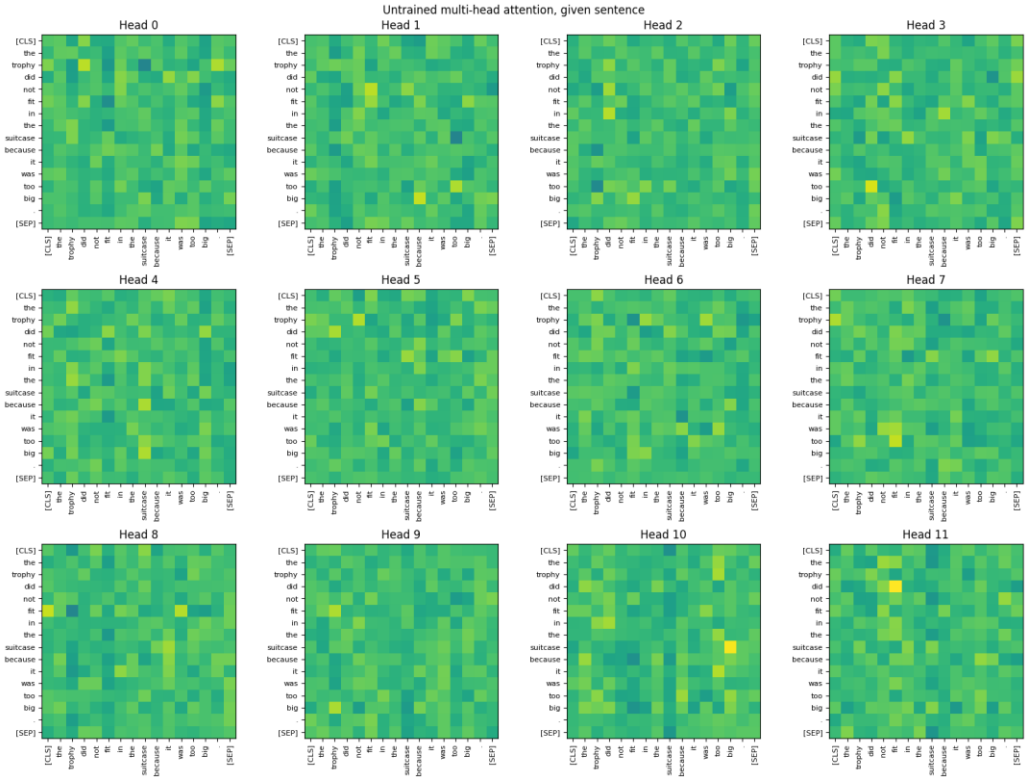
chef 0.275

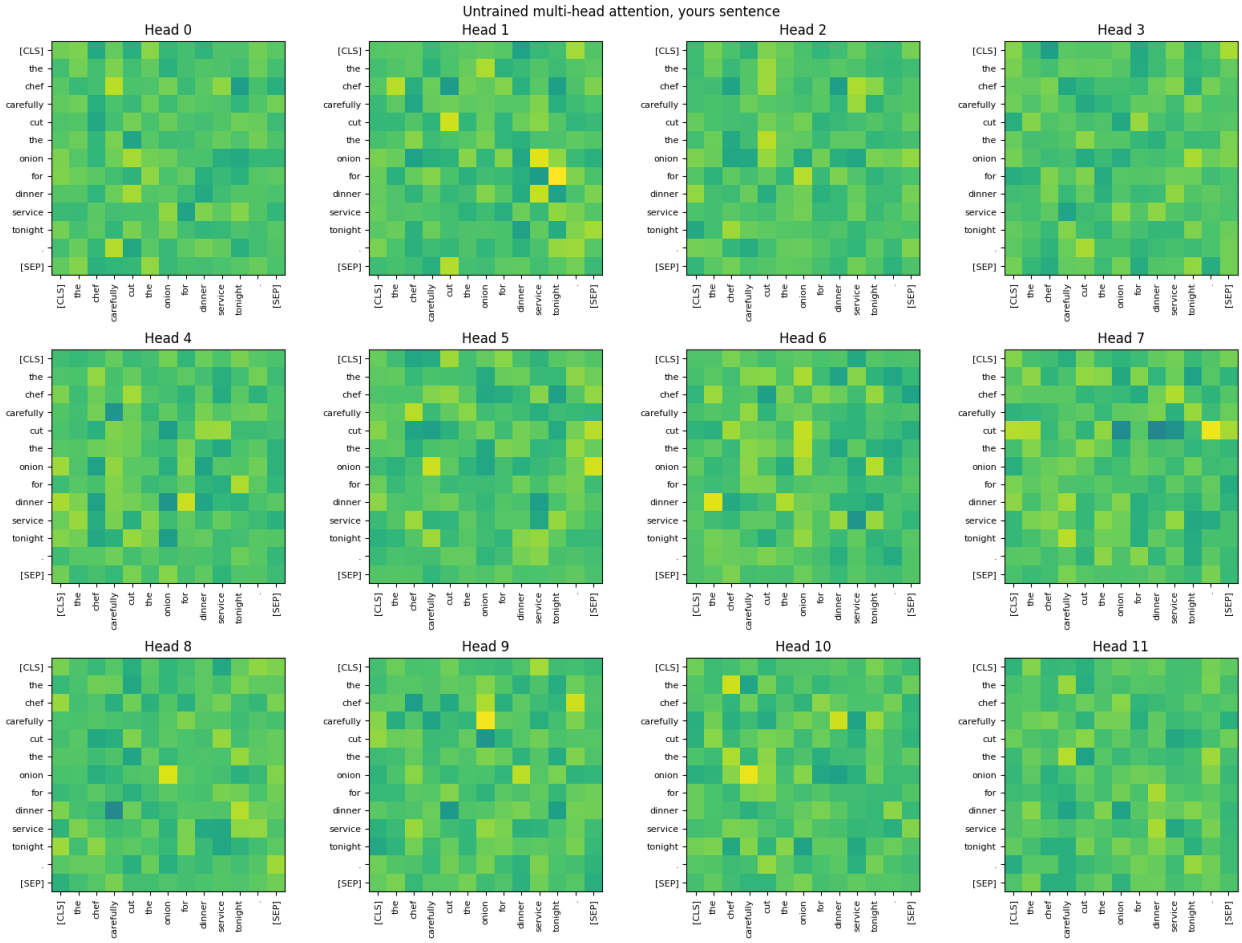
cook 0.212

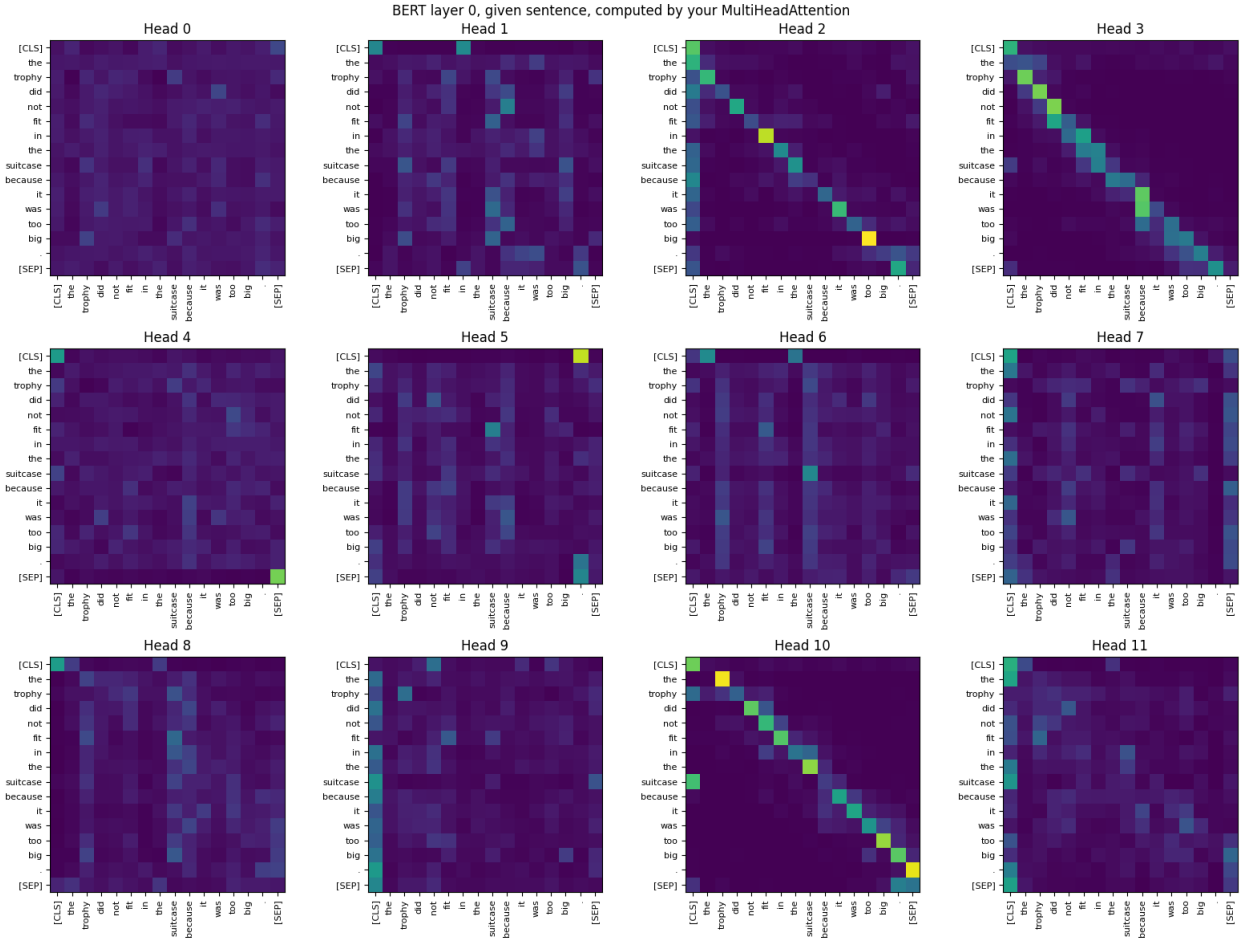
chefs 0.049

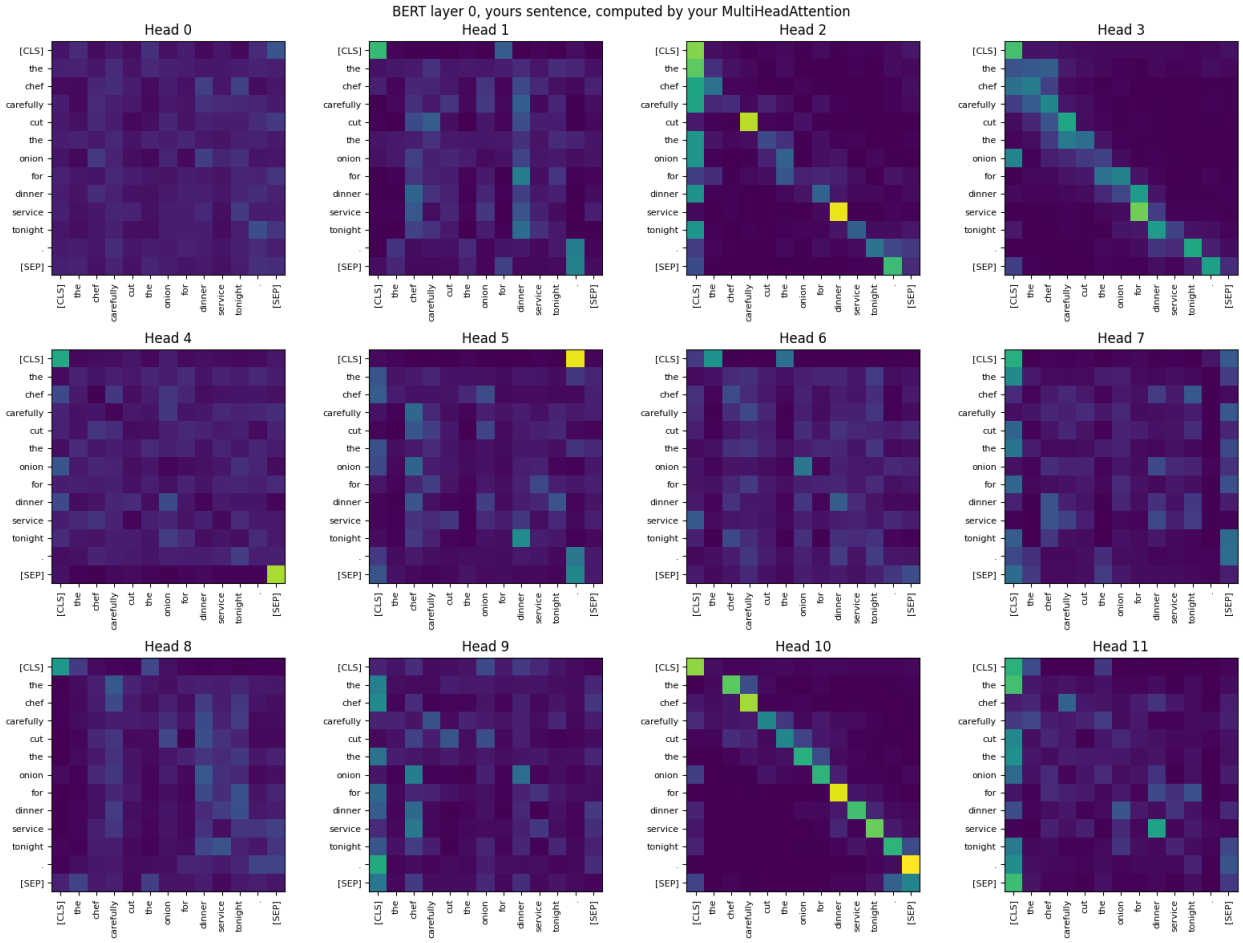
cooks 0.044

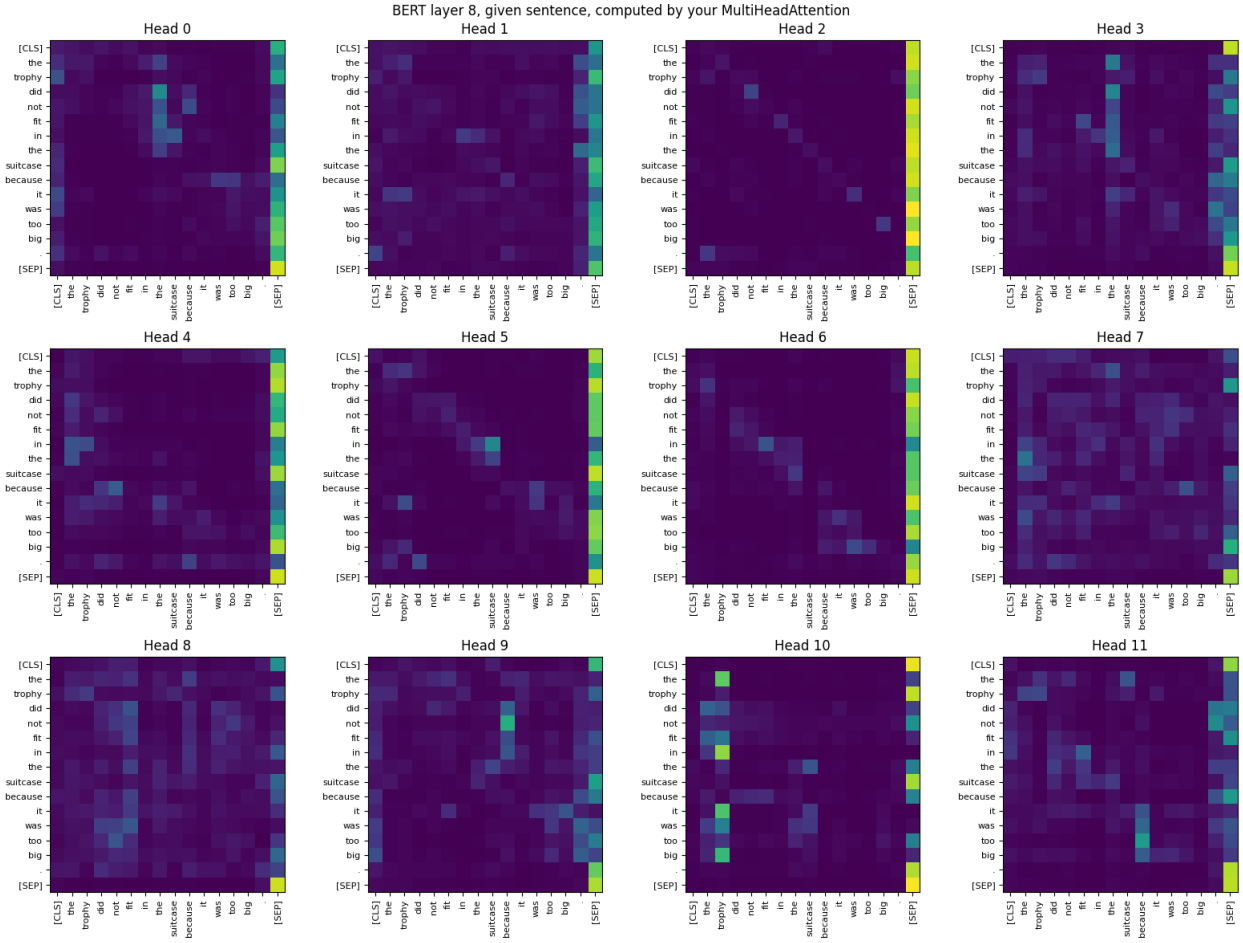
waiter 0.031

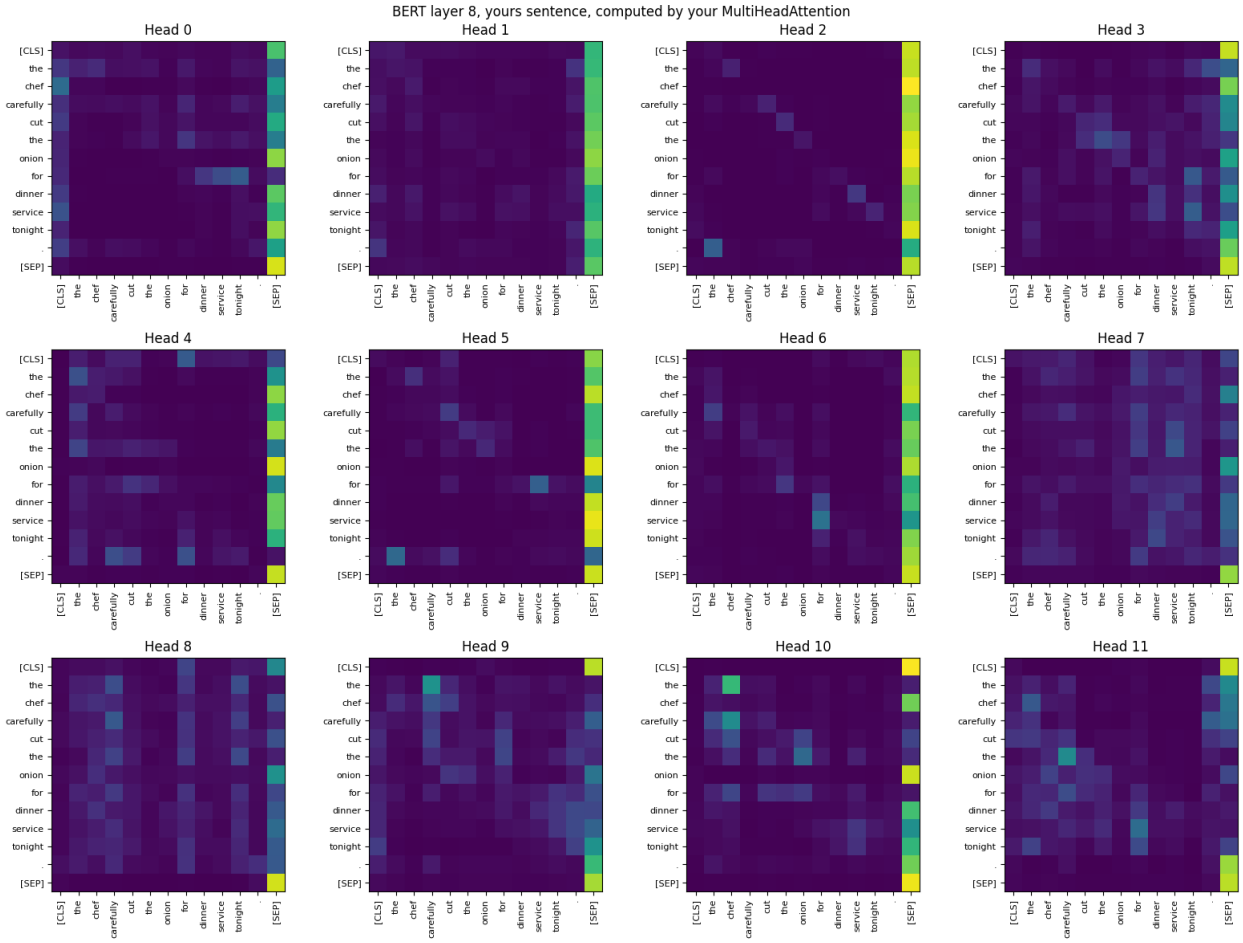



Untrained multi-head attention, yours sentence
Head 0
Head 1
Head 2
Head 3
Head 4
Head 5
Head 6
Head 7
Head 8
Head 9
Head 10
Head 11
[CLS]
the
chef
carefully
cut
the
onion
for
dinner
service
tonight
.
[SEP]


BERT layer 0, given sentence, computed by your MultiHeadAttention
Head 0
Head 1
Head 2
Head 3
Head 4
Head 5
Head 6
Head 7
Head 8
Head 9
Head 10
Head 11
[CLS]
the
trophy
did
not
fit
in
the
suitcase
because
it
was
too
big
.
[SEP]


BERT layer 0, yours sentence, computed by your MultiHeadAttention
Head 0
Head 1
Head 2
Head 3
Head 4
Head 5
Head 6
Head 7
Head 8
Head 9
Head 10
Head 11
[CLS]
the
chef
carefully
cut
the
onion
for
dinner
service
tonight
.
[SEP]


BERT layer 8, given sentence, computed by your MultiHeadAttention
Head 0
Head 1
Head 2
Head 3
Head 4
Head 5
Head 6
Head 7
Head 8
Head 9
Head 10
Head 11
[CLS]
the
trophy
did
not
fit
in
the
suitcase
because
it
was
too
big
.
[SEP]



BERT's guesses were pretty reasonable. Words like 'onion', 'dinner', and 'service' give clues. Looks like BERT used these words as heads for its guesses, with even the untrained heads focusing on 'dinner service'.